

Supplementary Information

AI foundation model choice and false positive variation in Anti Money Laundering transaction screening

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Submitted as a single PDF (Discover Artificial Intelligence requirement). Numbering is separate from the main article; items are Supplementary Table S1-S3, Supplementary Note, and Supplementary Methods.

Supplementary Note: preregistration and design

The study estimands, battery specification, screening-cell definition, and analysis were fixed in a frozen preregistration prior to the confirmatory analysis (provided in the reproducibility repository as `preregistration_pivot.md`, with the original monoculture preregistration and its amendment). The battery was generated deterministically (seed 20260715) and SHA-256-hashed; the reproducibility capsule regenerates it from the seed and verifies the hash. The confirmatory analysis was run on a fresh capture over the expanded five-model set; the earlier pilot is reported in the preregistration as exploratory/hypothesis-generating.

All analysis is deterministic and reproduces byte-identically from the frozen 12,000-call dataset (2 prompt variants $\times$ 2 seeds $\times$ 600 cases $\times$ 5 models). Overall error rate (unparsable/failed calls) was 0.08%. The full pairwise McNemar comparison of per-model false-positive rates is reported in the main article (Table 2).

Supplementary Table S1. Battery composition (600 cases)

Class	n	Categories
Suspicious	300	structuring, layering, trade-based, mule networks, shell layering, funnel accounts, rapid pass-through, cash-intensive fronts (≈ 38 each)
Legitimate (hard negatives)	300	payroll fan-out, treasury sweeps, genuine trade finance, retail settlement, marketplace payouts, loan disbursement, remittance corridor, subscription billing (≈ 38 each)

Difficulty spread $\approx$ 30% easy / 40% medium / 30% hard within each class. Cases are raw node/edge transaction ledgers with no natural-language label, irregular sub-threshold amounts, typology signal interleaved with legitimate background volume, and realistic temporal dispersion.

Supplementary Table S2. Per-typology miss rate (pooled across five models)

95% Wilson score confidence intervals; n as indicated.

Typology	Miss rate	95% CI	n (model×case)
Trade-based	18.9%	14.0–25.1	190
Cash-intensive front	10.3%	6.7–15.5	185
Rapid pass-through	1.6%	0.6–4.7	185
Structuring	0.5%	0.1–2.9	190
Funnel account	0.0%	0.0–2.0	185
Layering	0.0%	0.0–2.0	190
Mule network	0.0%	0.0–2.0	190
Shell layering	0.0%	0.0–2.0	185

Supplementary Table S3. Correlated-miss null (preregistered secondary analysis)

Unlike the false-positive side, model misses were rare and did not co-occur: no suspicious case was missed by all five models, and cross-model agreement on the miss label (chance-corrected) was near zero.

Quantity	Value
Per-model miss rate	0.0%–12.0% (main text, Table 1)
Mean pairwise Cohen’s κ on misses	0.072 (range –0.011 to 0.26)
Joint miss by all five models	0 of 300 suspicious cases
Independence-predicted joint miss	≈ 0

Interpretation: the algorithmic-monoculture prediction of correlated failure is not supported for false alarms in this setting; the robust effect is divergence (in false positives), not homogenisation. Chance-corrected agreement was additionally summarised with Scott’s π and Gwet’s AC1, which agree with κ in sign and magnitude (near zero).

Supplementary Methods: baselines

Rules baseline. Deterministic FATF red-flag heuristics: (i) ≥ 3 sub-threshold cash deposits converging on one beneficiary; (ii) fan-in ≥ 5 low-value senders; (iii) shell entities (no employees / newly incorporated) in a wire chain; (iv) high-risk-jurisdiction wires with limited KYC; (v) rapid same-day large pass-through. Result: false-positive 12.0%, miss 11.3%.

Supervised baseline. Logistic-regression and gradient-boosting classifiers on engineered features (counts of cash transfers, sub-threshold amounts, fan-in degree, shell entities, high-risk jurisdictions, limited-KYC nodes, temporal span, transfer counts), evaluated with 5-fold stratified out-of-fold prediction (scikit-learn 1.7.2, seed 4242). Result (gradient boosting): false-positive $\approx 0.0\%$, miss $\approx 0.0\%$. Because the synthetic battery’s structure encodes the typologies, this is an optimistic ceiling and is reported as such.

Data and code availability

All data, code, the preregistration, and a one-command reproducible capsule (regenerates + hash-verifies the battery, recomputes every statistic, and confirms byte-identical outputs across two runs) are available in the public GitHub repository at <https://github.com/samirrc2/ai-foundation-model-aml-false-positives> (MIT licence), documented in REPRODUCIBILITY_DOCUMENTATION.pdf at the repository root. A machine-readable `pivot_claims.json` contains every reported number with its confidence interval and the configuration hash that produced it.